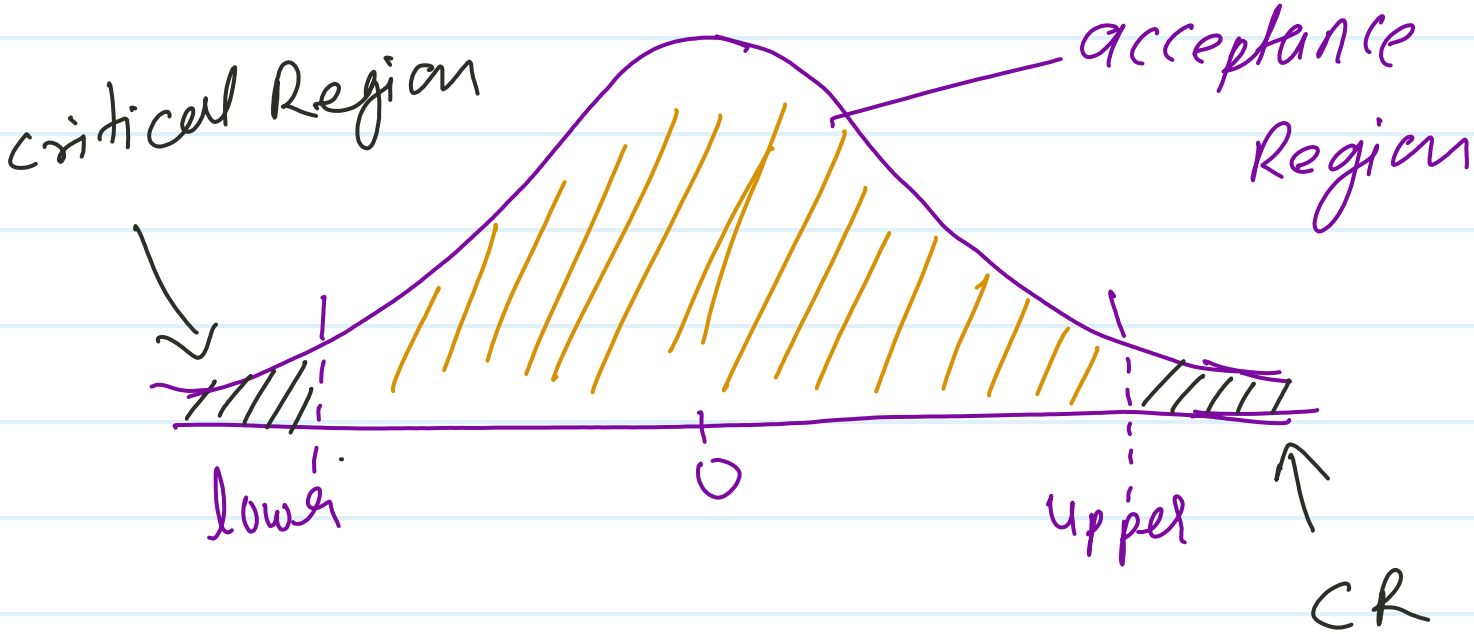
Hypothesis testing

* Confidence interval -



H_0 (null hypothesis) It support always confidence Interval. (Acceptance Region)

H_1 (Alternet hypothesis) It support always critical Region.

H_0 - We fail to reject H_0

H_1 = we reject H_0, and accept H_1.

$N = 10000$

$\mu = 285$

$n = 40$

$\bar{x} = 140$

Two tail -

$\mu = \bar{x}$ = Null hypothesis

$\mu \neq \bar{x}$ = Alternet hypothesis.

α = Critical Region

C.I = Acceptance Region

P = It is probability of H_0 to be True.

$P = 0.15$

$\alpha = 0.05$

$P > \alpha$

In this case we fail to reject null hypothesis.

$P < \alpha$

We reject null hypothesis and accept alternet hypothesis.

Type of hypothesis test

(i) Parametric test

(ii) Non-parametric test

① Parametric test

- (I) Z-test
- (II) T-test
- (III) Binomial test
- (IV) poisson test
- (V) exponential test

② Non-parametric test

- (I) chi-square test
- (II) Anova (F1)

{ Sample size < 30 & σ missing } T-test

{ Sample size ≥ 30 & σ given } Z-test.

Hypothesis testing

H_0 — Null hypothesis

H_1 — Alternet hypothesis

C.I.

critical / α

Doctor claim — 80% effective

$\Rightarrow \boxed{8/10}$

p value

Alternet hypothesis — to be true of null hypothesis